Letter

Metric-induced edge-dominated spectra in hyperbolic trusses

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Edge states in lattices typically rely on defects, topology, or parameter tuning, and occupy narrow spectral windows. Here, we demonstrate an edge-dominated mechanism driven by the Poincaré conformal metric. By projecting the hyperbolic metric onto elastic trusses, we show that the spatially varying bar lengths naturally embed the metric into the structural dynamic properties. This spatial modulation of the local dynamic scale inherently drives higher-index eigenmodes toward the boundary, yielding an accumulation of edge states. Furthermore, conformal mapping extends this mechanism to arbitrary boundaries, enabling the design of interconnected boundary-channel networks. Transient wave propagation confirms boundary-guided transport, establishing a topology-free route for programmable wave localization and energy focusing in mechanical metamaterials.

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Introduction. Wave and lattice dynamics have been extensively formulated in Euclidean geometry, where symmetry and Bloch's theorem determine the spectral structure [1–3]. Within this framework, edge modes and transport are mainly realized through two routes. The first is the topological mechanism, in which topologically protected edge states emerge within band gaps through nontrivial topological invariants and the bulk-edge correspondence [4–7]. The second is real-space modulation, where wave fields are confined at prescribed locations by means of defects [8–10], interface engineering [11,12], and parameter gradients [13–15]. These strategies have established versatile design paradigms for waveguiding and energy localization in artificial structures [16–18].

Recent work has extended lattice physics to non-Euclidean spaces, especially hyperbolic geometry with constant negative curvature [19–22]. Hyperbolic lattices exhibit a distinctive connectivity: the number of sites grows exponentially with graph distance [23,24]. This property has motivated studies of type-I [25–29] and type-II hyperbolic lattices [30–32], which have enabled advances in hyperbolic band theory [33–38], hyperbolic Chern insulators [39–42], hyperbolic quantum electrodynamics [43,44], and hyperbolic photonic topological insulators [45–47]. More recently, hyperbolic lattices have been realized in elastic wave systems, where edge states are sparsely accumulated in the spectra [48–51]. However, the fundamental physical origin of these edge-state spectra remains elusive. More importantly, how to encode the continuous Poincaré metric into discrete mechanical couplings to deterministically achieve edge-dominated spectra remains an open challenge.

Here, we demonstrate that the intrinsic metric of hyperbolic space can be elegantly embedded into elastic trusses, generating high-frequency spectra dominated by edge states. By establishing a mapping between the continuous metric and discrete lattice couplings, we reveal that metric-induced spatial inhomogeneity drives higher-index eigenmodes to the boundary in a scalar mass-spring model. We then extend this mechanism to vectorial elastic trusses, where the spatial modulation of physical bar lengths naturally encodes the Poincaré metric into the system's dynamic matrices. Finally, by conformal mapping, we unlock programmable network transport. These findings unveil a purely metric-driven route to edge localization, providing a new design paradigm for programmable waveguiding and energy-focusing metamaterials.

Edge-dominated spectra in Poincaré disks and rings. We begin with a truncated Poincaré disk, which is defined as $\mathcal{D} = \{z = re^{i\theta} | 0 \leqslant r \leqslant R_D,\ 0 \leqslant \theta < 2\pi,\ \mathrm{i} = \sqrt{-1}\}$ [Fig. 1(a)]. It is a conformal model of the hyperbolic plane with metric $\mathrm{d}s^2_{\mathrm{disk}} = 4(\mathrm{d}r^2 + r^2\mathrm{d}\theta^2)/(1-r^2)^2$ [52]. The divergence of this metric near $r = 1$ makes the boundary infinitely far away in hyperbolic distance. The wave spectral properties in hyperbolic space are governed by the Laplace-Beltrami eigenvalue problem [53] $(\Delta_g + \lambda_D)u(r,\theta) = 0$ with the free boundary condition. Here, $\Delta_g = \sigma_{\mathrm{I}}^{-1}(r)\Delta$, $\sigma_{\mathrm{I}}(r) = 4/(1-r^2)^2$ denotes the type-I conformal factor, λ_D is the eigenvalue, and Δ is the Euclidean Laplacian. To expose the localization mechanism, we introduce the hyperbolic radial coordinate $\varsigma_D = \ln[(1+r)/(1-r)]$. Then, the radial equation takes a Schrödinger-like form with effective potential $V_{\mathrm{eff}}^{\mathrm{disk}} = (\mu^2 - 1/4)/\sinh^2\varsigma_D$, $\mu \in \mathbb{Z}$. The radial dependence is given by $V_{\mathrm{disk}}^{\star} = 1/\sinh^2\varsigma_D$ [Fig. 1(a), inset], which is singular at the

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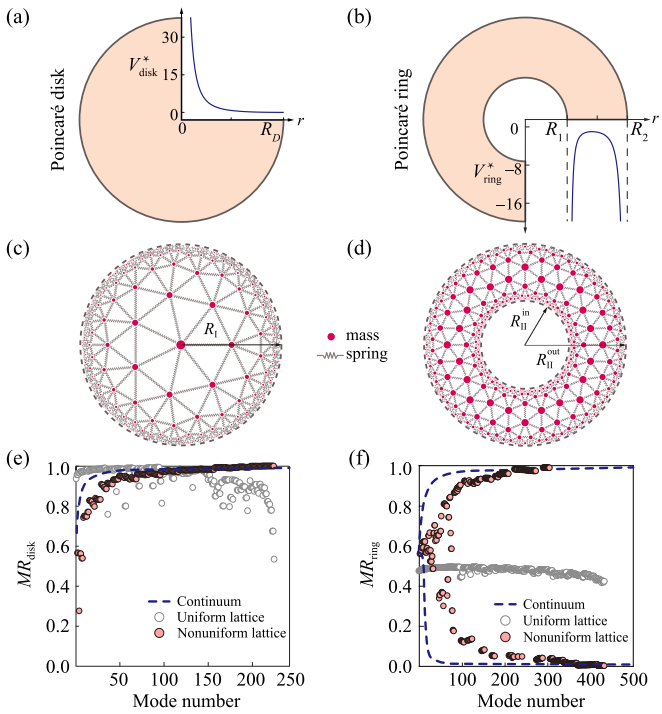


FIG. 1. Highly aggregated edge states in the hyperbolic continuum and discrete lattices. (a) Poincaré disk with $R_D = 0.999$. Inset: effective potential $V^{\star}_{\rm disk}$. (b) Poincaré ring with $R_1 = 0.427$ and $R_2 = 0.999$. Inset: effective potential $V^{\star}_{\rm ring}$. (c) A $\{3, 7\}$ type-I hyperbolic mass-spring lattice with $R_{\rm I} = 0.973$. (d) A $\{3, 7\}$ type-II hyperbolic mass-spring lattice with $R^{\rm in}_{\rm II} = 0.440$ and $R^{\rm out}_{\rm II} = 0.970$. (e) and (f) $MR_{\rm disk}$ and $MR_{\rm ring}$ for type-I and type-II lattices, respectively.

disk center and rapidly decreases toward the ideal boundary. It therefore suppresses the modal weight near the center and drives high-$|\mu|$ modes toward the outer region, giving rise to edge-localized eigenmodes, whose amplitudes decay inward from the boundary [Supplemental Material (SM) [54]].

To introduce an additional localization channel, we further consider the Poincaré ring, defined as $\mathcal{R} = \{f = re^{i\theta} | R_1 \leqslant r \leqslant R_2, 0 \leqslant \theta < 2\pi\}$, with $R_1 = 0.427$, $R_2 = 0.999$, and $r_h = 3.69$ [Fig. 1(b)]. Its metric is defined as $ds^2_{\rm ring} = r_h^2(dr^2 + r^2 d\theta^2)/[r^2 \sin^2(r_h \ln r)]$. This ring can be obtained from the disk through conformal mappings, which transfer the metric divergence of the disk boundary to both the inner and outer boundaries of the ring (SM [54]). For the ring $\Delta_g = \sigma_{\rm II}^{-1}\Delta$, $\sigma_{\rm II}(r) = r_h^2/[r^2 \sin^2(r_h \ln r)]$ represents the type-II conformal factor. Substituting this into the eigenvalue equation, we obtain the effective potential $V^{\rm ring}_{\rm eff}(\varsigma_R) = -\lambda_R/\sin^2\varsigma_R$ (SM [54]), where $\varsigma_R = r_h \ln r$, and λ_R is the eigenvalue. Thus, the radial trend is governed by $V^{\star}_{\rm ring}(\varsigma_R) = -1/\sin^2\varsigma_R$ [Fig. 1(b), inset]. Unlike the disk potential, this potential diverges to negative infinity at both radial boundaries, forming two attractive edge wells.

Metric-induced edge accumulation in mass-spring systems. Hyperbolic lattices are regular tessellations of negatively curved space, in which the number of boundary sites grows exponentially with graph distance (SM [54]). To bridge the continuum mechanism and discrete realizations, we construct type-I and type-II hyperbolic lattices from a $\{3, 7\}$ tessellation [Figs. 1(c) and 1(d)], where the Schläfli symbol $\{p, q\}$ denotes regular p-gon with q polygons meeting at each vertex. We first consider a uniformly coupled scalar mass-spring network with nodal mass m_0 and spring stiffness k_0. This model is commonly used as a low-order discretization of a hyperbolic continuum, where the graph Laplacian approximates the Laplace-Beltrami operator in the long-wavelength limit [55,56]. For an out-of-plane spring system, the Hamiltonian can be expressed as

$$H = \sum_i \varepsilon_i c_i^\dagger c_i + \sum_{\langle i,j \rangle} h_{ij}(c_i^\dagger c_j + c_j^\dagger c_i), \tag{1}$$

where $c_i^\dagger$ and c_i are formal site-basis operators associated with the displacement degree of freedom (DOF) at site i, $\varepsilon_i = k_0 d_i/m_i$, $h_{ij} = -k_0/\sqrt{m_i m_j}$, and d_i is the coordination number of site i.

In uniform coupling models, hyperbolic geometry dictates the lattice connectivity [55]. While the geometric proliferation of boundary sites inherently increases the DOF near the edge, the local coupling strength remains strictly constant. To rigorously compare radial localization of continuum and discrete eigenmodes, we evaluate the *mean modal radius*. For Poincaré disk, it is defined as

$$MR_{\rm disk} = \frac{\int_{\Omega_{\rm disk}} \eta_{\rm disk} |u(x,y)|^2 dA}{\int_{\Omega_{\rm disk}} |u(x,y)|^2 dA}, \quad \eta_{\rm disk} = \frac{r}{R_D}, \tag{2}$$

where $r = \sqrt{x^2 + y^2}$, R_D denotes the radius of the disk, and $\Omega_{\rm disk}$ is the truncated disk domain. $MR_{\rm disk} \to 1$ indicates that the mode is concentrated near the boundary.

For Poincaré ring, it can support modes localized near either the inner or the outer boundary. To distinguish these two edge modes, we define

$$MR_{\rm ring} = \frac{\int_{\Omega_{\rm ring}} \eta_{\rm ring} |u(x,y)|^2 dA}{\int_{\Omega_{\rm ring}} |u(x,y)|^2 dA}, \quad \eta_{\rm ring} = \frac{r - R_1}{R_2 - R_1}, \tag{3}$$

where R_1 and R_2 denote the inner and outer radii of the ring, respectively. When $MR_{\rm ring} \to 0$ ($MR_{\rm ring} \to 1$), it denotes inner-edge (outer-edge) localization. As shown in Figs. 1(e) and 1(f), as the mode order increases, $MR_{\rm disk}$ tends to 1 and $MR_{\rm ring}$ tends to either 0 or 1, implying strong localization at the disk and ring boundary. The corresponding radial amplitude decays inward from the boundary (SM [54]). It indicates that the negatively curved hyperbolic metric produces a macroscopic accumulation of edge states in the eigenvalue spectrum, which is distinct from previous cases where curvature acts as a geometric potential for real-space topological defects [57].

Next, we evaluate the discrete *mean modal radii*, $MR_{\rm disk}$ and $MR_{\rm ring}$, by replacing the continuum integrals with sums over the individual lattice sites (SM [54]). The resulting spectra reveal that both type-I and type-II uniform lattices deviate from their continuum counterparts, failing to capture the boundary-accumulation characteristic of higher-index modes [white dots in Figs. 1(e) and 1(f)]. Although $MR_{\rm ring}$ remains close to 0.5 throughout the spectrum, it reflects distinct modal distributions: low-order modes localize simultaneously at the inner and outer boundaries, whereas high-order modes become predominantly bulklike and depart from the continuum

trend (SM [54]). Consequently, mimicking the hyperbolic connectivity alone is insufficient; capturing the true higher-index continuum behavior strictly requires spatial modulation to directly govern the local lattice dynamics.

To implement this spatial modulation and restore the continuum behavior, we vary the discrete lattice mass according to the hyperbolic metric $m_i = m_0\sigma^{-1}(r_i)$, where $\sigma^{-1} = \sigma_{\mathrm{I}}^{-1}$ for the disk and $\sigma^{-1} = \sigma_{\mathrm{II}}^{-1}$ for the ring. This is because, under the conformal projection, discrete hyperbolic cells with equal area $\mathrm{d}A_g$ map onto the Euclidean plane with local areas $\mathrm{d}A_E$ proportional to the inverse conformal factor, $\mathrm{d}A_E = \sigma^{-1}\mathrm{d}A_g$. Consequently, maintaining a uniform physical mass density requires the lumped nodal mass to scale with $m_i = m_0\Delta A_{E,i}/\Delta A_{g,i} = m_0\sigma^{-1}(r_i)$. The dynamic matrix acquires the onsite energies $\varepsilon_i \approx \sigma(r_i)k_0 d_i/m_0$ and hopping amplitudes $h_{ij} \approx -\sigma(r_{ij})k_0/m_0$. Because $\sigma^{-1}(r)$ decreases toward the boundary, the resulting metric gradient reorganizes the spectrum and drives high-frequency modes to the boundaries (SM [54]). The resulting *mean modal radius* distributions now closely follow the continuum trend [pink points in Figs. 1(e) and 1(f)]. In contrast to uniformly coupled hyperbolic graphs, this edge accumulation is driven mechanically by the embedded geometry—via metric-induced mass modulation—rather than relying on topology or the simple exponential scaling of boundary sites.

Elastic trusses modulated by the hyperbolic metric. While the mass-modulated scalar lattice successfully recovers the continuum's edge states, artificially tuning individual parameters throughout a discrete system is highly impractical and cumbersome. This raises a critical question: can a physical architecture *naturally* embed this hyperbolic metric and its resulting edge-state accumulation effect without requiring tedious manual parameter tuning? The vectorial elastic truss provides an elegant solution [Figs. 2(a) and 2(b)]. In this physical realization, scalar displacements are replaced by in-plane displacement vectors at each hinge, $\mathbf{u}_i = [u_{ix}, u_{iy}]^{\mathrm{T}}$, with each connecting bar supporting purely axial deformation along its specific orientation [Fig. 2(a), right panel]. Assuming the truss is constructed from uniform constituent members—sharing a constant Young's modulus E, mass density ρ_t, and cross-sectional area A_t—the mechanical properties of the entire system are dictated strictly by the varying Euclidean lengths of the individual bars, l_{ij}.

For a bar connecting nodes i and j with length l_{ij}, the axial stiffness is $k_{ij} = EA_t/l_{ij}$ and the bar mass is $m_{ij} = \rho_t A_t l_{ij}$. Thus, a single geometric parameter—the projected bar length—simultaneously governs both mechanical properties. Shorter bars are intrinsically stiffer and lighter, effectively elevating the local mechanical frequency scale. Crucially, the radial distribution of these lengths can directly mirror the conformal factors of the Poincaré geometries [Figs. 2(a) and 2(b)]. Specifically, in the type-I hyperbolic lattice (the disk), the normalized bar length scales as $\hat{l}_{ij} \sim \hat{l}^{\mathrm{I}}_{ij}$, where $l^{\mathrm{I}}_{ij} = 1-r^2_{ij}$, and r_{ij} is the radial position of the bar's midpoint [Fig. 2(c)]. For the type-II lattice (the ring), the length follows $\hat{l}_{ij} \sim \hat{l}^{\mathrm{II}}_{ij}$ [Fig. 2(d)], governed by $l^{\mathrm{II}}_{ij} = r_{ij}|\sin(r_h \ln r_{ij})|$. By exploiting this spatial variation, the hyperbolic metric is naturally embedded into the structural mass and stiffness matrices of the truss, establishing a purely geometric physical foundation for higher-index edge localization.

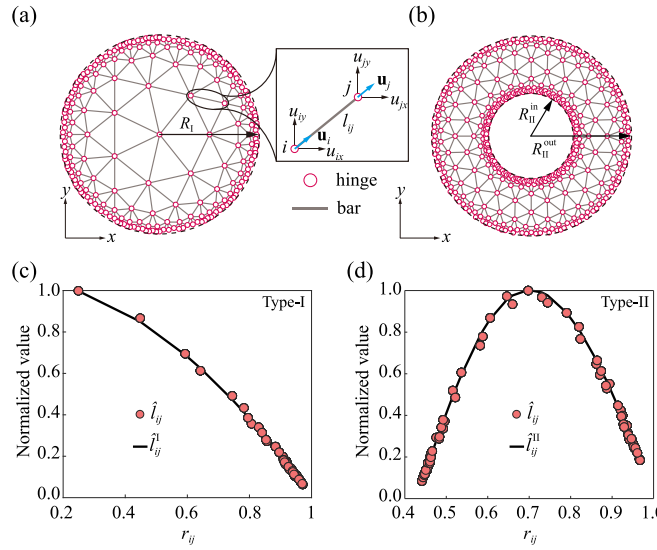


FIG. 2. Type-I and type-II hyperbolic trusses. (a) A {3, 7} type-I hyperbolic truss lattice with $R_{\mathrm{I}} = 0.973$. Right panel: force diagram of a truss member. (b) A type-II hyperbolic lattice with $R^{\mathrm{in}}_{\mathrm{II}} = 0.440$ and $R^{\mathrm{out}}_{\mathrm{II}} = 0.970$. (c) $\hat{l}_{ij}$ and $\hat{l}^{\mathrm{I}}_{ij}$ as functions of r_{ij} in the type-I hyperbolic truss lattice. (d) $\hat{l}_{ij}$ and $\hat{l}^{\mathrm{II}}_{ij}$ as functions of r_{ij} in the type-II hyperbolic truss lattice.

Metric-induced edge-state accumulation. Once the truss lengths are modulated by hyperbolic metrics, we find that a bar-scale parameter naturally emerges within the dynamic equations, reorganizing the system's eigenspectrum. We define the bar-scale parameter as

$$\chi_{ij} = \frac{m_{ij}}{k_{ij}} = \frac{\rho_t}{E} l^2_{ij}. \tag{4}$$

Substituting this relation into the mass-normalized stiffness matrix yields the following Hamiltonian (SM [54]):

$$\mathbf{H} = \sum_i \mathbf{c}^\dagger_i \varepsilon_i \mathbf{c}_i + \sum_{\langle i,j\rangle} (\mathbf{c}^\dagger_i \mathbf{h}_{ij}\mathbf{c}_j + \mathbf{c}^\dagger_j \mathbf{h}_{ji}\mathbf{c}_i), \tag{5}$$

where $\varepsilon_i = (1/m_i)\sum_{j\in\mathcal{N}_i} A_t\sqrt{\rho_t E/\chi_{ij}}\mathbf{P}_{ij}$, $\mathbf{h}_{ij} = -A_t\sqrt{\rho_t E/(\chi_{ij} m_i m_j)}\mathbf{P}_{ij}$, $\mathcal{N}_i$ denotes the set of nodes connected to node i, $\mathbf{c}_i = (c_{i,x}, c_{i,y})^T$, $\mathbf{n}_{ij}$ is the unit vector along the bar, $\mathbf{P}_{ij} = \mathbf{n}_{ij}\mathbf{n}^T_{ij}$ is a projector, and $m_i = (1/2)\sum_{j\in\mathcal{N}_i}\rho_t A_t l_{ij}$ is the lumped mass.

We apply a local radial approximation ($r_i \approx r_j \approx r_{ij}$) and normalize the material constants ($\rho_t = A_t = E = 1$). Then, the onsite energies $\varepsilon_i \sim (2/d_i)\sum_{j\in\mathcal{N}_i}\sigma(r_{ij})\mathbf{P}_{ij}$ and hopping amplitudes $\mathbf{h}_{ij} \sim -(2/\sqrt{d_i d_j})\sigma(r_{ij})\mathbf{P}_{ij}$ explicitly demonstrate that the hyperbolic metric dictates the dynamic behavior. Based on the designed bar lengths, the bar-scale parameter χ_{ij} inherently scales with the type-I (II) hyperbolic conformal factor $\chi_{ij} \sim \sigma^{-1}_{\mathrm{I(II)}}(r_{ij})$ [Figs. 3(a) and 3(b)].

The underlying physical mechanism is intuitively captured by the local frequency scale $\omega^2_{\mathrm{local}} = k/m = 1/\chi_{ij}$. Because the bars become progressively shorter (both stiffer and lighter) toward the boundary, this local frequency diverges radially outward, which drives high-frequency modes toward the boundary region. Consequently, despite the complexities of vectorial coupling, the macroscopic elastic truss perfectly

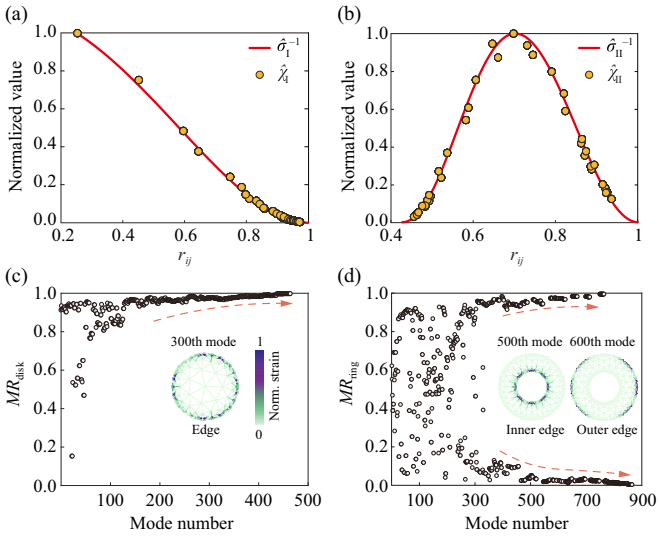


FIG. 3. Metric-induced edge-state accumulation in hyperbolic trusses. (a) The truss parameter $\hat{\chi}_{\mathrm{I}}$ (orange circles) and the inverse conformal factor $\hat{\sigma}_{\mathrm{I}}^{-1}$ (red curve) as functions of the bar-center radius r_{ij} for the type-I lattice. $\widehat{(\cdot)}$ denotes normalization by the maximum value. (b) Same as (a) but for the type-II lattice. (c) MR_{disk} versus mode number for the type-I lattice, showing progressive outer-boundary localization at higher mode orders. (d) MR_{ring} versus mode number for the type-II lattice; the branches approaching 0 and 1 correspond to localization near the inner and outer boundaries, respectively. Insets in (c) and (d) show the eigenmodes.

preserves the edge-localization mechanism of the scalar model. Mean radius evaluations MR_{disk} and MR_{ring} confirm this progressive outward shift and edge-state accumulation effect for higher-order modes [Figs. 3(c) and 3(d)]. Other kinds of conformal scaling χ can also sustain this effect as long as χ decreases sufficiently toward the boundary (SM [54]). Furthermore, the metric-induced edge localization persists across arbitrary $\{p, q\}$ tessellations. This localization arises from geometric scaling, and transient pulse simulations (SM [54]) verify that the metric-modulated trusses support boundary-guided wave transport, thereby providing a versatile structural platform for elastic energy focusing. However, this mechanism is neither topologically protected [58] nor induced by local parameter tuning [59]; the eigenmode distributions exhibit a degree of robustness against geometric defects (SM [54]).

Conformal generalization for programmable network transport. To demonstrate the potential of these structures for elastic waveguiding and energy concentration, we extend the edge-localized response of the type-I hyperbolic truss to alternative boundary geometries. Following the Riemann mapping theorem [60], any simply connected planar domain is conformally equivalent to the unit disk. Leveraging this principle, we apply specific conformal transformations $w = T_\beta(z)$, $\beta = 1, 2, 3$ to map a Poincaré disk lattice onto a triangle, a square, and a hexagon [Fig. 4(a)], where w denotes the coordinate in the transformed two-dimensional (2D) plane. These conformal maps are governed by the Schwarz-Christoffel formula [61]

$$T_\beta(z) = C_\beta \int_0^z (1 - \zeta^{\alpha_\beta})^{-2/\alpha_\beta} d\zeta, \tag{6}$$

where $\alpha_1 = 3$, $\alpha_2 = 4$, and $\alpha_3 = 6$ correspond to the triangular, square, and hexagonal mappings T_1, T_2, and T_3, respectively, and C_β is a normalization constant that sets the size of the mapped polygon.

These mappings change the global boundary contour while preserving local angles. The 2D normalized member parameter $\hat{\chi}_{ij}$ retains the spatial trend of the type-I lattice, with smaller values near the boundary and larger values in the bulk [Fig. 4(a)]. The induced metric remains conformal,

$$ds_\beta^2 = \frac{4\left|\left(T_\beta^{-1}(w)\right)'\right|^2}{\left(1 - \left|T_\beta^{-1}(w)\right|^2\right)^2}|dw|^2 = \sigma_\beta(w)|dw|^2, \beta = 1, 2, 3, \tag{7}$$

which preserves the boundary divergence of the hyperbolic metric (SM [54]). Because the mapped polygonal domains lack radial symmetry, σ_β now depends on both x and y. For each bar ij, we evaluate $\sigma_{\beta,ij}^{-1} = \sigma_\beta^{-1}(w_{ij})$ at its midpoint and normalize both quantities. Their agreement, $\hat{\chi}_{ij} \sim \hat{\sigma}_{\beta,ij}^{-1}$, confirms metric embedding [insets in Figs. 4(b)–4(d)]. This demonstrates that altering shape does not disrupt the underlying physics. In all three transformed geometries, the *mean modal radius MR* tends toward 1 as mode order increases (SM [54]), confirming that edge-dominated spectra persist independent of the specific global boundary contour [Figs. 4(b)–4(d)].

This boundary-shaping capability unlocks a direct route for designing programmable elastic waveguides. Using the conformal square geometry, we assemble unit cells into square-A and square-B configurations [insets of Figs. 4(e) and 4(f)] to construct interconnected boundary-channel networks. When excited by a Gaussian pulse at a designated boundary site, the transient wave remains tightly confined to these engineered channels, demonstrating highly controllable boundary-guided transport. To rigorously quantify this confinement, we evaluate the boundary-channel energy fraction $\Xi_*(t) = (\sum_{i\in\Gamma_*} |\mathbf{u}_i(t)|^2)/(\sum_{i\in\Omega_*} |\mathbf{u}_i(t)|^2)$, where $*$ indexes square-A or square-B, Γ_* denotes the boundary of the lattice domain Ω_*. As shown in Figs. 4(e) and 4(f), $\Xi_*(t)$ remains consistently high throughout the dynamic response, confirming that the boundary nodes dominate the energy transport. The transient amplitude fields [Figs. 4(g) and 4(h)] visually verify that the elastic wave traverses the designed network paths. Ultimately, these networks demonstrate how metric-induced edge-state accumulation can be actively harnessed for localized energy concentration and elastic waveguide design.

Conclusions and discussion. In summary, we have established a purely geometry-driven paradigm for generating dense edge-state spectra in elastic structures. By embedding the Poincaré conformal metric into the bar length of a vectorial truss, we have demonstrated that the spatial modulation of the local dynamic scale drives higher-index eigenmodes toward the boundary. This mechanism departs from conventional approaches that rely on topological invariants, localized defects, or parameter tuning. Furthermore, we have extended

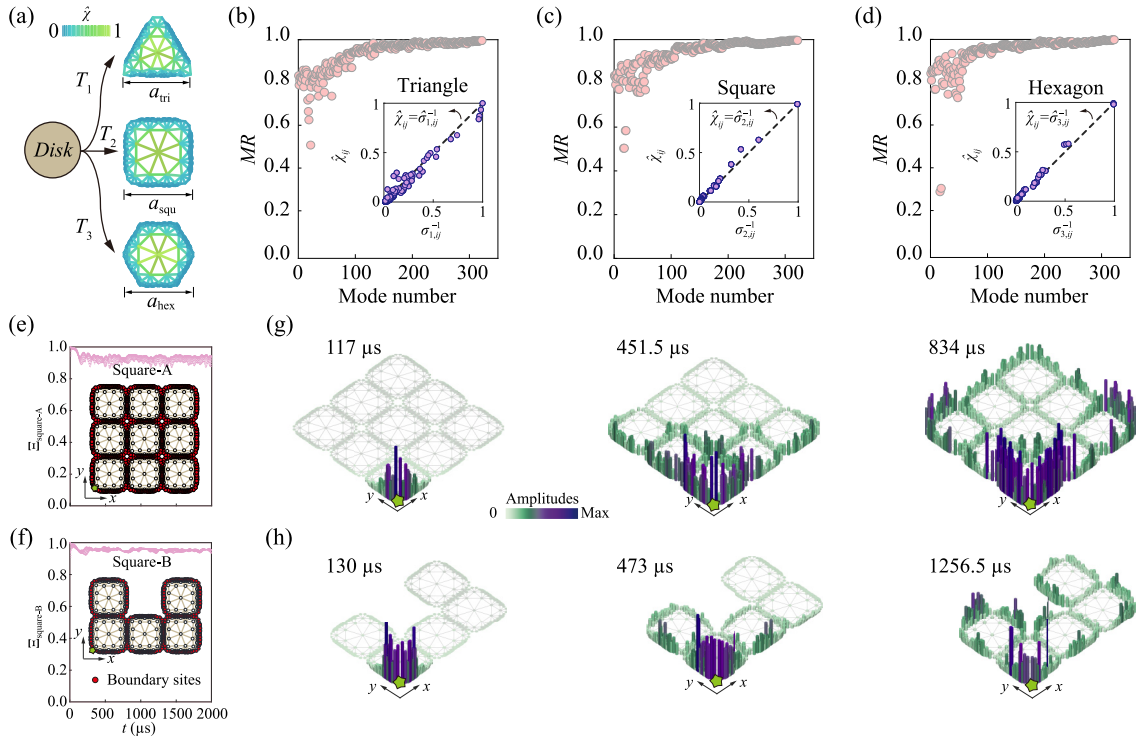


FIG. 4. Hyperbolic trusses under conformal mapping and network transmission. (a) Conformal mappings from the Poincaré disk to triangular, square, and hexagonal {3, 8} hyperbolic trusses, with $a_{\mathrm{tri}} = a_{\mathrm{squ}} = a_{\mathrm{hex}} = 0.2$ m. The color scale represents the spatial distribution of the $\hat{\chi}$. (b)–(d) MR distributions versus mode number for the triangular, square, and hexagonal trusses, respectively. Insets: comparison between the normalized $\hat{\chi}_{ij}$ and the normalized inverse conformal factor $\hat{\sigma}^{-1}_{\beta,ij}$. The black dashed line denotes $\hat{\chi}_{ij} = \hat{\sigma}^{-1}_{\beta,ij}$. (e) and (f) are the time-dependent boundary-channel energy fractions $\Xi(t)$ for the square-A and square-B lattices, respectively. Insets: the boundary-channel nodes and excitation sites. The excitation signals are $U\sin(2\pi f_2 t)e^{-t_3(t-t_4)^2}$ for square-A and $U\sin(2\pi f_3 t)e^{-t_5(t-t_6)^2}$ for square-B. $U = 1$ mm, $f_2 = 77.837$ kHz, $f_3 = 100.475$ kHz, $t_3 = 10^9$ s^{-2}, $t_5 = 5 \times 10^8$ s^{-2}, and $t_4 = t_6 = 10^{-4}$ s. Simulations use a time step of 0.5 μs and a total duration of 2000 μs. (g) and (h) are transient amplitude fields for square-A and square-B, respectively.

this edge-dominated effect to arbitrary boundaries via conformal mapping and realized programmable network transport. Overall, translating these curved-space metrics into mechanical architectures establishes new possibilities for elastic wave manipulations, paving the way for advanced vibration isolation and phononic devices.

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Data availability. The data that support the findings of this article are not publicly available upon publication because it is not technically feasible and/or the cost of preparing, depositing, and hosting the data would be prohibitive within the terms of this research project. The data are available from the authors upon reasonable request.

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